REGULATORY REPORTING DATA LAYER

A common data layer for regulatory and risk reporting in banks, participation banks and investment firms

Start with an assessment of the reporting chain, build the data layer per report family, then run it through every reporting period

1 **The challenge**
Reporting challenges, what the data layer is, BCBS 239 principles. p. 2 – 4

2 **How it is built**
Architecture, data quality and validation, report families. p. 5 – 7

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Three tiers, the assessment, delivery approach, customer inputs. p. 8 – 11

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Regulatory Reporting Challenges

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Regulatory reports are produced on a fixed calendar, from many source systems, under supervisory review. The issues below usually sit in the data behind the reports, not in the reports themselves.

01 · THE CHALLENGE

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Many reports, many data paths

- Each report family has its own extract
- The same figure differs between reports
- One change is made in several places

Report inventory

Manual steps at period end

- Excel files between extract and submission
- Adjustments made by hand, not recorded
- Dependency on a few people

Manual work

Findings from the regulator or audit

- BDDK / TCMB or audit remarks on data
- Corrections and resubmissions
- Evidence asked for each figure

Findings

Period-end delays

- Deadlines met with overtime
- Late source data holds up all reports
- Little time left for checks

Timeliness

No traceable path to the source

- Lineage is known by people, not documented
- Reconciliation to the GL is partial
- The impact of a change is not known in advance

Lineage

Regulation keeps changing

- New reports, fields and formats
- IFRS 9 and risk data needs grow
- Core banking migration changes the sources

Change

WHAT A COMMON DATA LAYER CHANGES

One data layer
shared by all report families

Reconciled to the GL
before any report is built

DQ rules before submission
thresholds and a correction loop

Lineage source → report
documented per report

Audit trail
who changed what, and when

What the Regulatory Reporting Data Layer Is

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A service package that builds one harmonised, reconciled and traceable data layer for regulatory and risk reports. Report families are added one at a time on a common foundation.

02 · WHAT IT IS

FIVE REPORT FAMILIES

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- **TCMB / BDDK legal reports**
  prudential and statistical reporting
- **IFRS 9 (TFRS 9)**
  staging, ECL inputs, collateral, cash flows
- **BCBS 239 / PERDAR**
  risk data aggregation, lineage and controls evidence
- **Credit risk reporting**
  exposure, rating, collateral, limits, concentration
- **EBA reports (optional)**
  for banks with an EU parent or an EU subsidiary

FOR WHOM

Banks, participation banks and investment firms that report to BDDK and TCMB. «DIGITALLENCY DOĞRULASIN»

A data layer, not a reporting tool
The package builds the data that feeds the reports: harmonised, reconciled to the GL, checked and traceable. Report layouts and submission software stay as they are.

WHEN IT IS USEFUL

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- BDDK / TCMB findings or audit remarks on reporting data
- A new regulation or a change in report format
- Reporting that depends on manual Excel steps
- Delays at period end
- Data issues in IFRS 9 model inputs
- A core banking migration

WHAT IT IS NOT

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- Cleansing data in the source systems
- Report layout design or submission software
- Correspondence with the regulator
- Legal interpretation of regulations (the institution decides)
- Changes to core systems
- An enterprise data governance framework (separate service)

BCBS 239 Principles and What Changes

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BCBS 239 sets out principles for risk data aggregation and risk reporting. The package uses them as the backbone: the assessment measures each principle group, the data layer builds what is missing.

03 · BCBS 239

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PRINCIPLE GROUP	WHAT THE ASSESSMENT MEASURES (A)	WHAT THE DATA LAYER BUILDS (B)	EVIDENCE
Governance and infrastructure P1 governance · P2 data architecture and IT infrastructure	Ownership of risk data; how the reporting chain is built; manual steps and tools	Common data model; data owners per data area; documented architecture	Data model, owner list, architecture document
Risk data aggregation P3 accuracy · P4 completeness · P5 timeliness · P6 adaptability	Accuracy, completeness and timeliness for priority reports; effort for an ad hoc request	GL reconciliation, DQ rules and completeness checks; scheduled runs; reusable data demand layer	Reconciliation results, DQ rule catalogue and results, run times
Risk reporting P7 – P11	Report inventory, frequency, owners, manual steps	The data that feeds the reports; report layouts are out of scope	Lineage from source to report
Supervisory review P12 – P14	Not assessed: these are the supervisor's responsibilities	—	The evidence above supports the review

STATUS PER PRINCIPLE IN THE ASSESSMENT

Met | Partly met | Not met | Not assessed

PERDAR

In Türkiye, PERDAR is treated as the BDDK counterpart of the BCBS 239 principles; the same structure is used for it.

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Architecture

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A generic, platform-independent architecture. Source data is harmonised once in a common layer and reconciled to the GL; each report family reads report-ready data from its own data demand layer.

04 · ARCHITECTURE

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Source systems

where the data is created

- Core banking
- Loans and limits
- Treasury
- Collateral
- GL and rating

Staging

copies of source data, as delivered

- Loaded on the reporting calendar
- Load checks
- Kept for audit

Common regulatory data layer

harmonised, historised, reconciled to the GL

- Transaction
- Contract / account
- Customer / counterparty
- Collateral
- GL

Data demand layer

report-ready datasets per report family

- TCMB / BDDK
- IFRS 9
- BCBS 239 / PERDAR
- Credit risk
- EBA (optional)

Reporting tools

and regulator formats

- Existing reporting tools
- Regulator file formats
- Internal risk reports

CROSS-CUTTING CONTROLS

DQ rules and validation

Rule catalogue and a rule-based validation engine; checks before submission.

Reconciliation controls

Totals reconciled to the GL and between layers.

Lineage source → report

Each report field traced back to its source.

Audit trail

Loads, corrections and manual adjustments recorded.

TOOL AND PLATFORM EXAMPLES · AGREED IN THE SCOPE

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Informatica

PowerCenter or IDMC for integration; Informatica Data Quality for rules.

Existing data warehouse

Teradata, Oracle or SQL Server, when the layer is built on the current DWH.

Cloud data platform

When reporting data moves to the institution's cloud platform.

«PLATFORM»

The platform and tools are chosen with the institution in the scope.

Data Quality and Validation

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Data is checked before any report is built. Rules are kept in one catalogue, run on every load, and results above an agreed threshold go back for correction.

05 · DQ AND VALIDATION

RULE TYPES IN THE CATALOGUE

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- **Completeness**
  mandatory fields filled, all records loaded
- **Validity and format**
  code lists, value ranges, formats
- **Consistency**
  same figure across reports and layers
- **Reconciliation to the GL**
  balances agree with the general ledger
- **Plausibility**
  unexpected changes against the last period
- **Timeliness**
  data arrived by the calendar cut-off

VALIDATION FLOW BEFORE SUBMISSION

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Data loaded
Source data reaches staging and the common layer.

→

Rules run
All rules for the report family run automatically.

→

Threshold check
Error rate compared with the agreed threshold «NUMBER».

→

Results recorded
Results per rule, report and period are stored.

Within threshold
Data is released to the data demand layer and the report.

Above threshold
The data set is returned to the data owner for correction.

Corrected data is loaded again and the rules run again before submission.

Reconciliation controls
GL and layer totals compared on every run; differences explained.

DQ monitoring report
Rule results and trends per period for the reporting unit.

Thresholds are agreed, not assumed. Rules and thresholds are set with the regulatory reporting unit. The legal interpretation of a regulation stays with the institution.

Report Families

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Each report family is a module on the common layer. Typical sources, critical data elements, DQ rules and frequency are confirmed for each institution in the assessment.

06 · REPORT FAMILIES

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TCMB / BDDK legal reports

Prudential and statistical reporting

TYPICAL SOURCES

Core banking, loans, treasury, GL

CRITICAL DATA ELEMENTS

Balance, product type, counterparty sector, currency, maturity

TYPICAL DQ RULES

GL reconciliation, code list checks, completeness

FREQUENCY

Daily to quarterly, by report

IFRS 9 (TFRS 9)

Staging, ECL inputs, collateral, cash flows

TYPICAL SOURCES

Loans, collateral, rating, GL

CRITICAL DATA ELEMENTS

Stage, days past due, PD / LGD / EAD inputs, collateral value, contractual cash flows

TYPICAL DQ RULES

Stage consistency, missing ratings, collateral coverage

FREQUENCY

Monthly and quarterly

BCBS 239 / PERDAR

Risk data aggregation; lineage and controls evidence

TYPICAL SOURCES

All risk data sources through the common layer

CRITICAL DATA ELEMENTS

Critical data element list per risk report, with owners

TYPICAL DQ RULES

Accuracy, completeness and timeliness measures

FREQUENCY

Each reporting cycle

Credit risk reporting

Exposure, rating, collateral, limits, concentration

TYPICAL SOURCES

Loans, limits, rating, collateral, treasury

CRITICAL DATA ELEMENTS

Exposure, rating, limit, collateral, risk group

TYPICAL DQ RULES

Limit vs exposure, rating coverage, concentration totals

FREQUENCY

Daily and monthly

EBA reports

For banks with an EU parent or an EU subsidiary

TYPICAL SOURCES

Common layer and group mappings

CRITICAL DATA ELEMENTS

Fields of the EBA templates in scope «FIELD»

TYPICAL DQ RULES

EBA validation rules; consistency with local reports

FREQUENCY

Quarterly

OPTIONAL MODULE

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Three Service Tiers: Assessment, Data Layer, Run

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The assessment maps the current reporting chain. The data layer builds the common foundation and the report families. Run supports every reporting period and each regulation change.

07 · SERVICE TIERS

A — Regulatory Data Assessment

fixed fee «...» · 4 – 8 weeks

- Report inventory and manual steps
- Source → report lineage, priority reports
- BCBS 239 gap assessment per principle
- Data quality issues list
- Target architecture, roadmap, estimate

OUTPUT

Report inventory · lineage map · BCBS 239 gap table · DQ issues list · target architecture · data layer roadmap and estimate

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B — Regulatory Data Layer

fee «...» per report family · foundation once

- Common data layer model and build
- Data demand layer per report family
- DQ rule catalogue, validation engine
- Reconciliation, lineage documentation
- Parallel run and handover

OUTPUT

Report-ready data per family · active DQ rules · documented lineage · parallel run accepted

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C — Run

monthly fee «...» · minimum term 12 months

- Period-end run support, DQ result review
- Regulation changes from monthly capacity
- Monthly DQ monitoring report
- Small improvements
- Unused hours are not carried over

OUTPUT

Monthly report · periods closed on time · new reports, fields and rules from a monthly capacity

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Assessment first. The data layer starts from the assessment, or from an equivalent analysis already done by the institution. «DIGITALLENCY DOĞRULASIN»

Assessment: Work Flow and Timeline

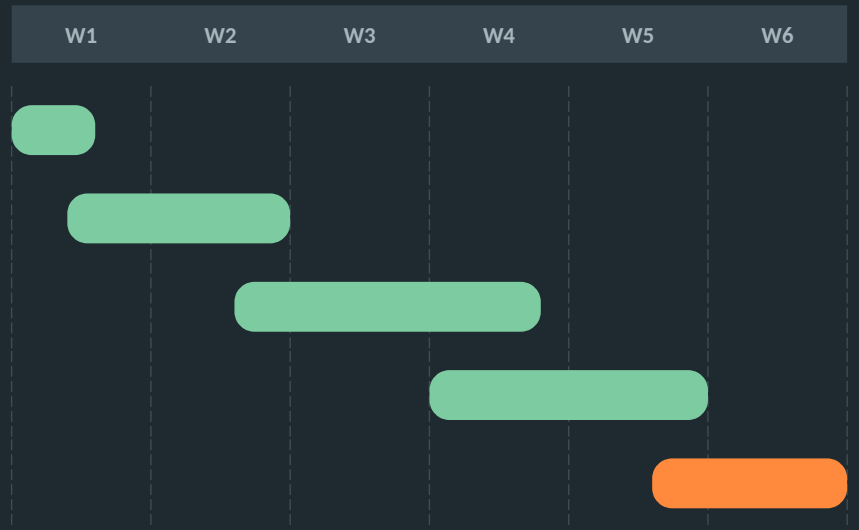
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Five phases over four to eight weeks. The assessment ends with a BCBS 239 status per principle, a target architecture and a roadmap for the data layer.

08 · ASSESSMENT

INDICATIVE PLAN · BAND M · 6 WEEKS

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BCBS 239 STATUS PER PRINCIPLE

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Met | Partly met | Not met | Not assessed

Each status is backed by evidence from the inventory, the lineage work and DQ checks on priority reports.

DELIVERABLES

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- Report inventory (reports, frequency, owners, tools, manual steps)
- Source → report lineage map for priority reports
- BCBS 239 gap table per principle
- Data quality issues list
- Target architecture
- Data layer roadmap and estimate

Duration by size. Small about 4, medium about 6, large about 8 weeks. Size depends on report count, source systems, reporting frequency and DQ rule count. «DIGITALLENCY DOĞRULASIN»

Data Layer Delivery and Parallel Run

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The common foundation is built once. Each report family is then added and run in parallel with the current reports until the results agree.

09 · DATA LAYER DELIVERY

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1 — Common layer model

Data areas, keys, history rules, GL mapping.

2 — Build and GL reconciliation

Loads from the sources; totals reconciled to the GL.

3 — Data demand layer

Report-ready datasets for the report family.

4 — DQ rules and validation

Rules, thresholds and the return loop active.

5 — Parallel run

New and current reports compared period by period.

6 — Handover

Runbook, lineage documents, training.

FOUNDATION · ONCE

PER REPORT FAMILY

HANDOVER

PARALLEL RUN

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- Current report
- Report from the data layer

→ **Compare**
field by field, each period

→ **Differences explained**
Each difference is traced to a source, a rule or the current process and closed with the reporting unit.

ACCEPTANCE

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- Results within the agreed tolerance «NUMBER» for each report family
- Tolerance met for «NUMBER» consecutive periods
- DQ rules active with documented thresholds
- Lineage documented for the reports in scope

Current reports keep running. The institution submits from the current process until the parallel run is accepted; the switch is planned per report family.

What the Institution Provides

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Regulatory reporting knowledge sits with the institution. The work needs time from three groups, and three conditions decide when it can start.

10 · CUSTOMER INPUTS

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Regulatory reporting unit

finance and risk

- Report list, calendar and owners
- Explanation of manual steps
- Decisions on rules and thresholds
- Review of parallel run results

Data owners

per source system

- Meaning of source fields
- Correction of returned data
- Approval of data definitions
- Contact for DQ issues

IT

infrastructure and access

- Read access to the sources
- Environments on the agreed platform
- Scheduling and operations
- Security and access approvals

CONDITIONS TO START

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Access to regulatory SMEs
Reporting specialists available on a planned schedule.

Access to the sources
Read access to source systems or agreed extracts.

GL data for reconciliation
GL balances at the level needed to reconcile the layer.

Interpretation stays with the institution. Komtaş builds and checks the data. How a regulation is read, and what is submitted, is decided by the institution's regulatory unit.

Experience

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Reference work on regulatory reporting data, shown by sector.

REFERENCE WORK

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Sector	Work	Scope	Year
A private bank	**Legal reporting data layer**	Common data layer for IFRS 9, VTM, PERDAR, EBA and credit risk reporting. «DIGITALLENCY DOĞRULASIN»	«YEAR»
A participation bank	**Legal reporting data in the EDW**	Legal reporting data moved from the core banking platform to the enterprise data warehouse.	«YEAR»
A credit bureau	**Validation of bank submissions**	Rule-based validation of data sent by banks; a submission above a 5% error threshold is returned to the bank. Built with Informatica.	«YEAR»

WHAT THE REFERENCES SHOW

One layer, several report families

IFRS 9, legal, risk and EBA reporting fed from the same data layer.

Reporting data in the warehouse

Legal reporting data moved off the core banking platform into the data warehouse.

Rules with a return loop

Submissions checked against rules and a threshold, and returned when they fail.

References are shown by sector, without customer names.

About Komtaş, and the Next Step

Who delivers the Regulatory Reporting Data Layer, and what is needed to start.

11 · ABOUT KOMTAŞ AND NEXT STEP

WHY KOMTAŞ

Google Cloud Premier partner
Komtaş is a Google Cloud Premier partner.

Microsoft Azure partner
Komtaş is also a Microsoft Azure partner.

Data warehouse and integration work in finance
Data warehouse, data integration and data quality projects for banks. «DIGITALLENCY DOĞRULASIN»

Regulatory reporting data
Reference work by sector on p. 12. «DIGITALLENCY DOĞRULASIN»

From assessment to monthly run
Assessment, data layer and Run from one partner.

Since 1989, focused on data · 250+ specialists · offices in Istanbul, Ankara, Izmir and Dubai

NEXT STEPS

1. **Walkthrough** — today — you are here
2. **Discovery call** — reports, findings, sources, platform
3. **Assessment proposal** — fixed fee, scope and plan
4. **Kick-off** — access, report inventory starts

Useful for the discovery call. The list of reports in scope and their frequency; the source systems and the current data warehouse; recent BDDK / TCMB findings or audit remarks; the current reporting tools and manual steps.